Overview of Ensemble learning

Subject: Ensemble learning

1.

Amended Cross-Entropy Cost: An Approach for Encouraging Diversity in Classification Ensemble The most common approach for training classifier is using Cross-entropy cost function. However, one would like to train an ensemble of models that have diversity so when we combine them it would provide best results. Assuming we use a simple ensemble of averaging K {\displaystyle K} classifiers. Then the Amended Cross-Entropy Cost is

2.

Medicine Ensemble classifiers have been successfully applied in neuroscience, proteomics and medical diagnosis like in neuro-cognitive disorder (i.e. Alzheimer or myotonic dystrophy) detection based on MRI datasets, cervical cytology classification. Besides, ensembles have been successfully applied in medical segmentation tasks, for example brain tumor and hyperintensities segmentation.

3.

The Bayes optimal classifier is a classification technique. It is an ensemble of all the hypotheses in the hypothesis space. On average, no other ensemble can outperform it. The Naive Bayes classifier is a version of this that assumes that the data is conditionally independent on the class and makes the computation more feasible. Each hypothesis is given a vote proportional to the likelihood that the training dataset would be sampled from a system if that hypothesis were true. To facilitate training data of finite size, the vote of each hypothesis is also multiplied by the prior probability of that hypothesis. The Bayes optimal classifier can be expressed with the following equation:

4.

Financial decision-making Ensemble learning methods have been widely adopted in finance for tasks such as credit scoring, bankruptcy prediction, and risk management. By combining multiple base models, ensembles can exploit nonlinear relationships, handle high-dimensional and noisy data, and often deliver more stable out-of-sample performance than single models or traditional statistical baselines such as logistic regression and linear factor models. This approach aligns with the broader trend in financial machine learning described by Marcos López de Prado, who argues that combining diverse learners can improve robustness, mitigate overfitting, and extract persistent patterns from noisy financial time series. In retail and corporate credit scoring, ensemble classifiers such as random forests, gradient boosting machines and stacked models are frequently used to assess default risk. A recent systematic literature review of machine-learning credit-scoring studies published between 2018 and 2024 reports that tree-based ensembles and boosting methods are among the most commonly applied techniques and typically achieve higher predictive accuracy than traditional scorecards or single classifiers. Ensemble methods have also been applied to corporate failure and bankruptcy prediction. Studies comparing different ensemble constructions (such as bagging, boosting and heterogeneous classifier pools) report that well-tuned ensembles tend to achieve higher classification accuracy and more robust performance across industries than individual models. The accuracy of prediction of business failure is a very crucial issue in financial decision-making. Therefore, different ensemble classifiers are proposed to predict financial crises and financial distress. Also, in the trade-based manipulation problem, where traders attempt to manipulate stock prices by buying and selling activities, ensemble classifiers are required to analyze the changes in the stock market data and detect suspicious symptom of stock price manipulation. Implementation of ensemble learning not only with machine learning models but also with mathematical frameworks such as hidden Markov models for applications in trading, regime detection, and asset pricing remains a rapidly developing area of research.

5.

Ensemble learning applications In recent years, due to growing computational power, which allows for training in large ensemble learning in a reasonable time frame, the number of ensemble learning applications has grown increasingly. Some of the applications of ensemble classifiers include:

6.

y = a r g m a x c j ∈ C ∑ h i ∈ H P (c j | h i) P (T | h i) P (h i) {\displaystyle y={\underset {c_{j}\in C}{\mathrm {argmax} }}\sum _{h_{i}\in H}{P(c_{j}|h_{i})P(T|h_{i})P(h_{i})}}

7.

While speech recognition is mainly based on deep learning because most of the industry players in this field like Google, Microsoft and IBM reveal that the core technology of their speech recognition is based on this approach, speech-based emotion recognition can also have a satisfactory performance with ensemble learning. It is also being successfully used in facial emotion recognition.

8.

Land cover mapping Land cover mapping is one of the major applications of Earth observation satellite sensors, using remote sensing and geospatial data, to identify the materials and objects which are located on the surface of target areas. Generally, the classes of target materials include roads, buildings, rivers, lakes, and vegetation. Some different ensemble learning approaches based on artificial neural networks, kernel principal component analysis (KPCA), decision trees with boosting, random forest and automatic design of multiple classifier systems, are proposed to efficiently identify land cover objects.

9.

Malware Detection Classification of malware codes such as computer viruses, computer worms, trojans, ransomware and spywares with the usage of machine learning techniques, is inspired by the document categorization problem. Ensemble learning systems have shown a proper efficacy in this area. Model Hardening Malware detection models, like all machine learning models, are vulnerable to adversarial machine learning attacks where an attacker pushes the boundary of what is classified as malware. By rotating through an ensemble of models using moving target defense, the attacker has a diminished knowledge advantage.

10.

Intrusion detection An intrusion detection system monitors computer network or computer systems to identify intruder codes like an anomaly detection process. Ensemble learning successfully aids such monitoring systems to reduce their total error.

11.

Ensemble theory Empirically, ensembles tend to yield better results when there is a significant diversity among the models. Many ensemble methods, therefore, seek to promote diversity among the models they combine. Although perhaps non-intuitive, more random algorithms (like random decision trees) can be used to produce a stronger ensemble than very deliberate algorithms (like entropy-reducing decision trees). Using a variety of strong learning algorithms, however, has been shown to be more effective than using techniques that attempt to dumb-down the models in order to promote diversity. It is possible to increase diversity in the training stage of the model using correlation for regression tasks or using information measures such as cross entropy for classification tasks.

12.

Stacking Stacking (sometimes called stacked generalization) involves training a model to combine the predictions of several other learning algorithms. First, all of the other algorithms are trained using the available data, then a combiner algorithm (final estimator) is trained to make a final prediction using all the predictions of the other algorithms (base estimators) as additional inputs or using cross-validated predictions from the base estimators which can prevent overfitting. If an arbitrary combiner algorithm is used, then stacking can theoretically represent any of the ensemble techniques described in this article, although, in practice, a logistic regression model is often used as the combiner. Stacking typically yields performance better than any single one of the trained models. It has been successfully used on both supervised learning tasks (regression, classification and distance learning) and unsupervised learning (density estimation). It has also been used to estimate bagging's error rate. It has been reported to out-perform Bayesian model-averaging. The two top-performers in the Netflix competition utilized blending, which may be considered a form of stacking. Bayesian predictive stacking generalizes the idea of stacking from the statistical estimation literature to the combination of posterior predictive distributions. These ideas have also been developed and investigated for Gaussian process models, especially for spatial data analysis and can be used to construct transfer learning frameworks for massive spatial data sets.